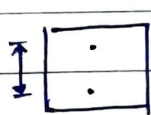
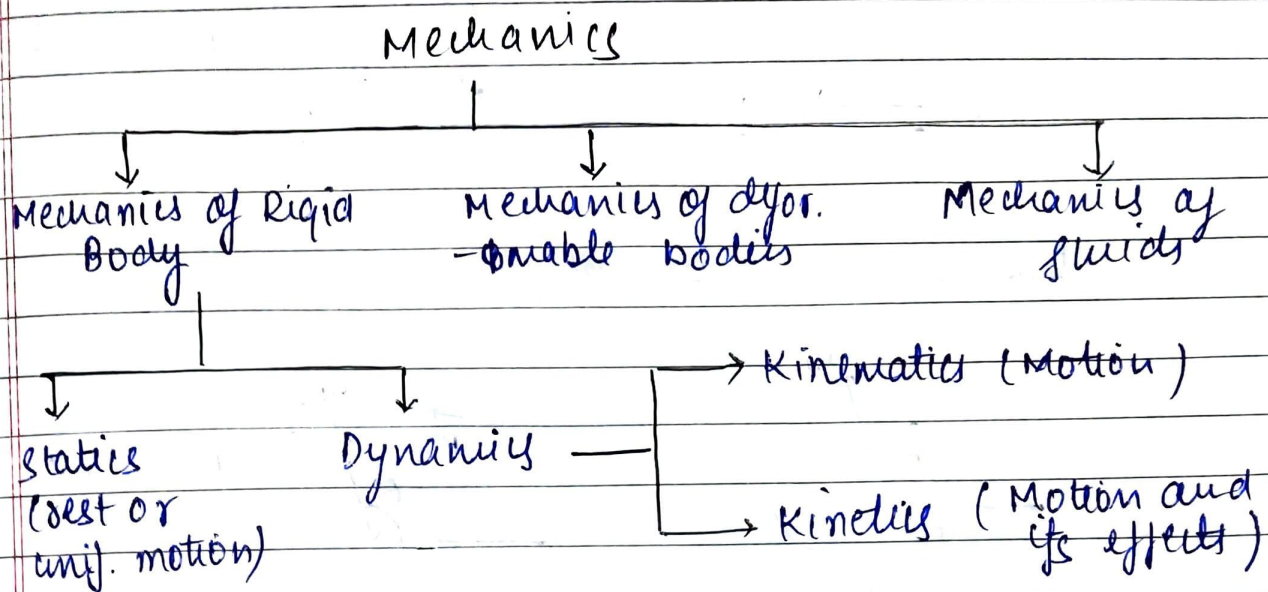
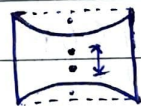


Introduction to Engineering Mechanics



For rigid bodies distance b/w two points is always same i.e. it does not deform



→ Deformable Bodies

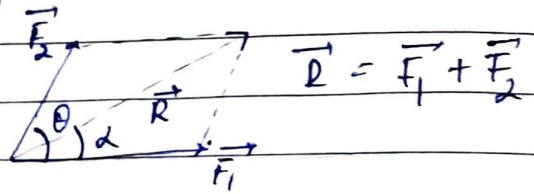
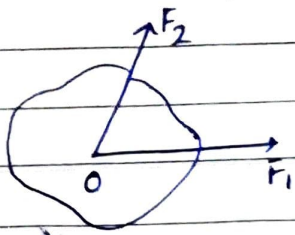
No body is completely rigid. A body where deformities is negligible is called rigid body.

- For rigid body we study external effect of forces.
- For deor. bodies we study internal effects such as stress, strain, strength etc.

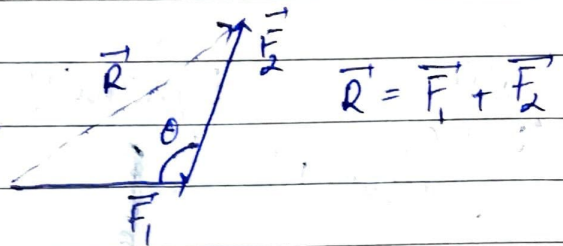
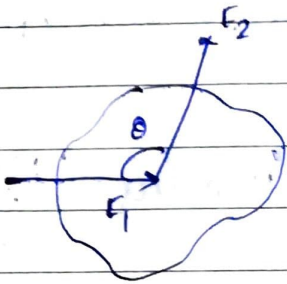
Fundamental Laws of Mechanics

- Newton's 1st Law of motion
- " 2nd " " "
- " 3rd " " "
- " Law of gravitation
- " Law of forces
- Δ Law of forces
- Principle of transmissibility of forces
- Law of superposition

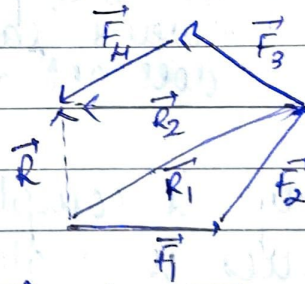
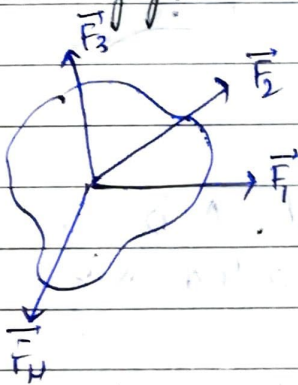
↳ Parallelogram Law of Forces



↳ Triangle Law of Forces



↳ Polygon Law of Forces

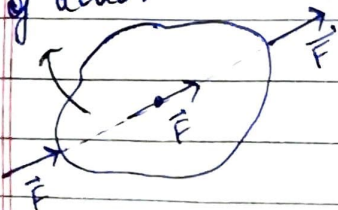


$$\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4$$

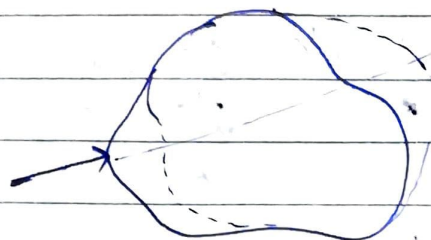
$$\vec{R} = \vec{R}_1 + \vec{R}_2$$

↳ Principle of Transmissibility of Force

line of action

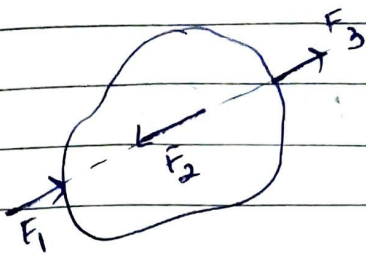


Deformable Bodies



On line of action force on a deformable body will differently deform at different position

→ Principle of superposition



$$\vec{R} = \vec{F}_1 + (-\vec{F}_2) + \vec{F}_3$$

System of Forces

coplanar (2D)

Non-coplanar (3D)

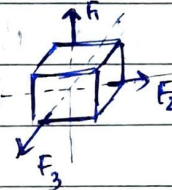
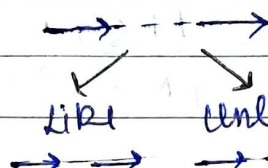
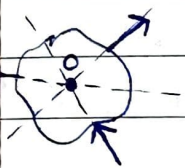
Concurrent

Non-concurrent

Collinear

Concurrent

Non-concurrent



Parallel

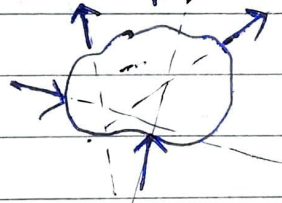
Non-Parallel

Parallel

Non-parallel

Like

unlike



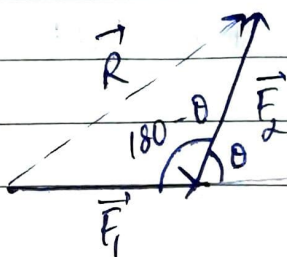
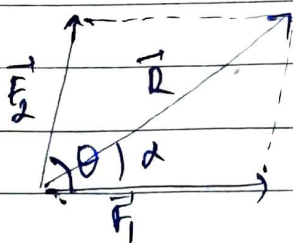
Like

unlike

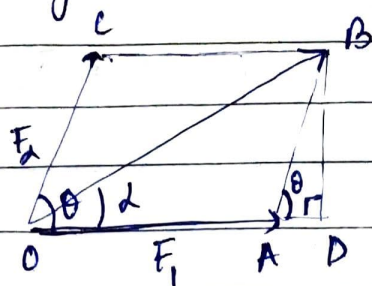
Comparison of two forces

Graphical Method } // sign and taking care of
Analytical Method } forces

Graphical method (using Protractor and Scale)



Analytical Method



$$\{ AD = AB \cos \theta \}$$

In ΔOBD

$$R^2 = OD^2 + BD^2$$

$$R^2 = (OA + AD)^2 + BD^2$$

$$R^2 = OA^2 + AD^2 + 2(OA)(AD) + BD^2$$

$$R^2 = OA^2 + AB^2 + 2(OA)(AD)$$

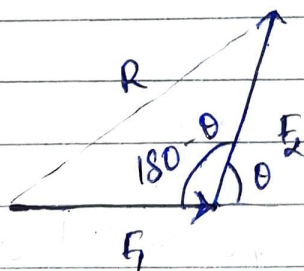
$$R^2 = F_1^2 + F_2^2 + 2F_1F_2 \cos \theta$$

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$

$$\tan \alpha = \frac{BD}{OD} = \frac{AB \sin \theta}{OA + AB \cos \theta} = \frac{F_2 \sin \theta}{F_1 + F_2 \cos \theta}$$

$$\alpha = \tan^{-1} \left(\frac{F_2 \sin \theta}{F_1 + F_2 \cos \theta} \right)$$

Triangle Law Method

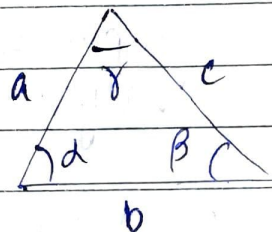


using cosine law:-

$$R^2 = F_1^2 + F_2^2 - 2F_1F_2 \cos(180 - \theta)$$

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$

cosine law



$$a^2 = b^2 + c^2 - 2bc \cos \beta$$

$$b^2 = a^2 + c^2 - 2ac \cos \gamma$$

Projection Method

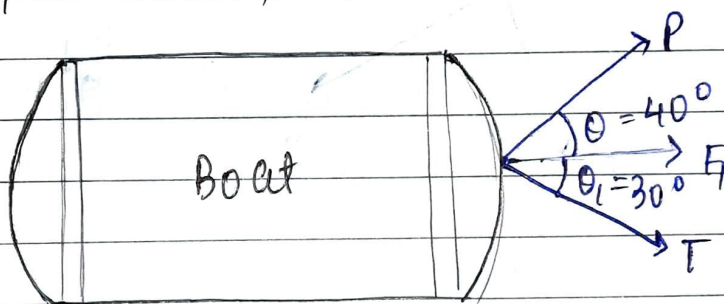
Resolve all the vectors in horizontal & vertical components and :

$$R_x = \sum F_x \quad R_y = \sum F_y$$

$$R = \sqrt{R_x^2 + R_y^2}$$

$$\tan \theta = \frac{R_y}{R_x}$$

Ques. The boat is to be pulled onto the shore using 2 ropes. Determine the magnitude of forces T & P acting in each rope in order to develop a resultant force $F_r = 100 \text{ kN}$ directed along the keel axis $a-a$ as shown. The angle $\theta = 40^\circ$, $\theta_1 = 30^\circ$.



$$R_x = 100 \text{ kN}, \quad R_y = 0$$

$$P \cos 40^\circ + T \cos 30^\circ = 100000$$

$$P(0.76) + T(0.86) = 100000$$

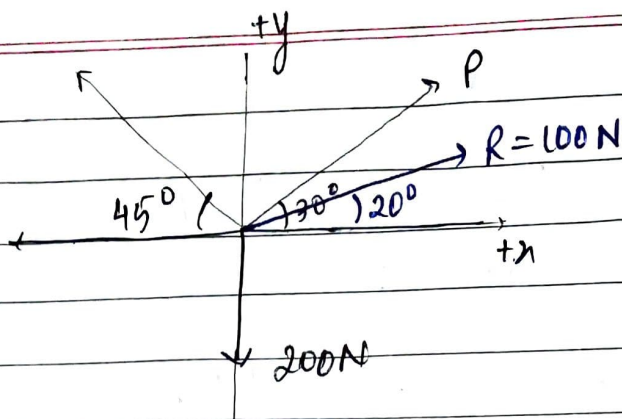
$$P \sin 40^\circ - T \sin 30^\circ = 0$$

$$P(0.64) = T(0.50)$$

$$P = T(0.78)$$

find P & T using above eqⁿ

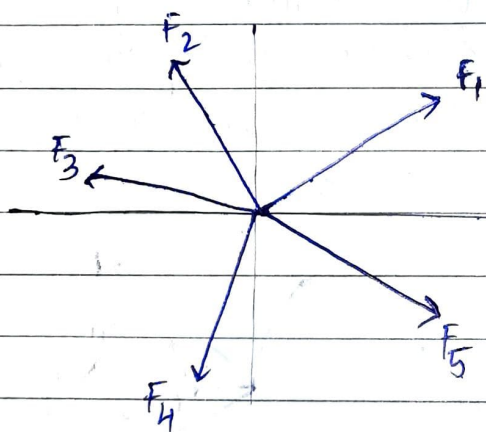
Ques. Determine the values of P and T , so that the resultant of the 3 coplanar forces is 100 N at 20° to the x -axis.



$$R_x = \sum F_x = P \cos 30 - Q \cos 45 = 100 \cos 20$$

$$R_y = \sum F_y = P \sin 30 + Q \cos 45 - 200 = 100 \sin 20$$

Equilibrium of Forces



condition for equilibrium:

$$R = 0$$

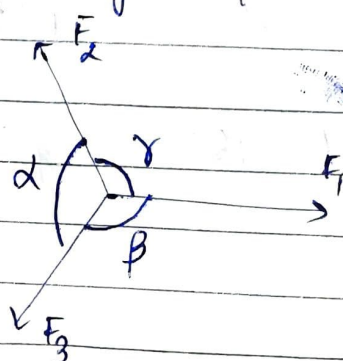
$$R_x = F_x = 0$$

$$R_y = F_y = 0$$

$$\sum M = 0$$

Lami's Theorem

If 3 coplanar forces acting at a point or particle be in equilibrium, then each force is proportional to the sine of angle b/w the other two forces.



Applicable for coplanar & concurrent vectors in an equilibrium condition

$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$$

Beam

Simple Supported Beam

one fixed other roller

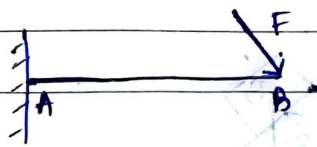


Roller support ϕ
(If support there is elongation in beam then to overcome the effect)



Cantilever Beam

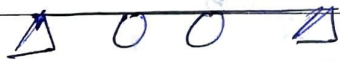
one fixed other free



Fixed Beam



Continuous Beam

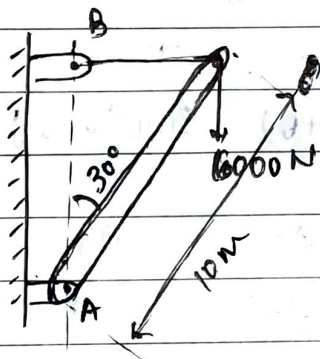


More than two supp.

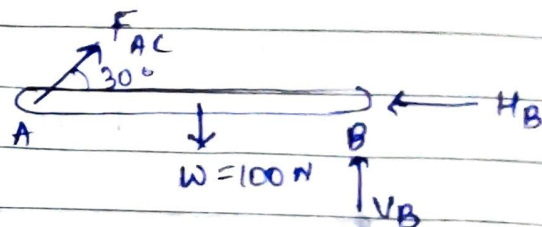
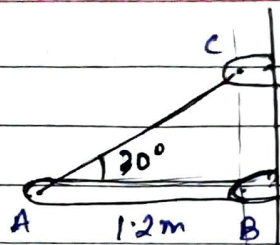
Overhanging Beam



Ques.

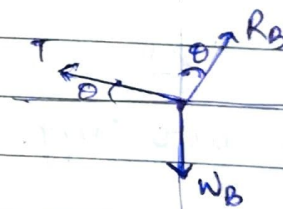
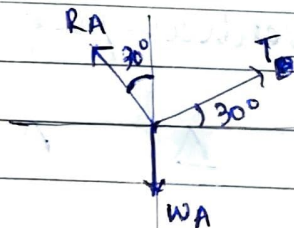
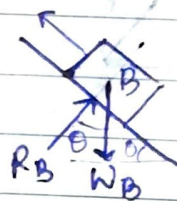
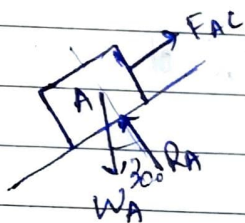
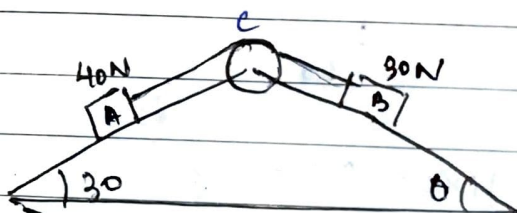


Ques.



$$\begin{aligned}\sum F_x &= 0 & H_B - F_{AC} \cos 30 &= 0 \\ \sum F_y &= 0 & V_B - 100 + F_{AC} \sin 30 &= 0 \\ \sum M_A &= 0 & W \times 0.6 - V_B \times 1.2 &= 0\end{aligned}$$

Ques.



$$\frac{R_A}{\sin 120} = \frac{T}{\sin 150} = \frac{W_A}{\sin 90}$$

$$\frac{R_B}{\sin (90 + \theta)} = \frac{T}{\sin (180 - \theta)} = \frac{W_B}{\sin 90}$$

$$\frac{R_A}{(\sqrt{3}/2)} = \frac{T}{(1/2)} = 40$$

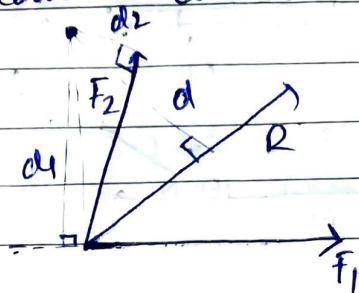
$$\frac{20}{\sin \theta} = \frac{30}{1}$$

$$T = 20 \text{ N}$$

$$\sin \theta = \frac{2}{3} \quad \theta = \sin^{-1}(2/3)$$

Varignon's Theorem (Principle of Moment)

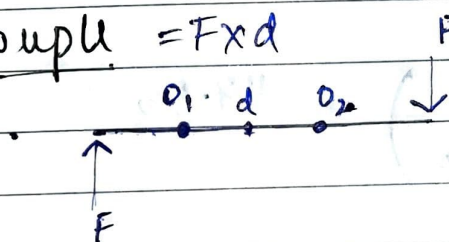
The algebraic sum of moments of a system of coplanar forces about a moment centre is equal to the moment of their resultant forces about the same moment centre.



Moment of resultant = Sum of moment of all force

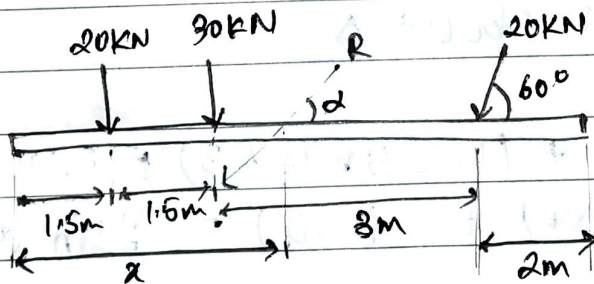
$$R \times d = F_1 d_1 + F_2 d_2$$

Couple = $F \times d$



Couple at O_1 and O_2 is same

Ans.



$$R_x = \sum F_x = -20 \cos 60^\circ = -10 \text{ kN}$$

$$R_y = \sum F_y = (-20 - 30 - 20 \sin 60^\circ) \text{ kN} = -67.32 \text{ kN} \\ = 67.3 \text{ kN} \downarrow$$

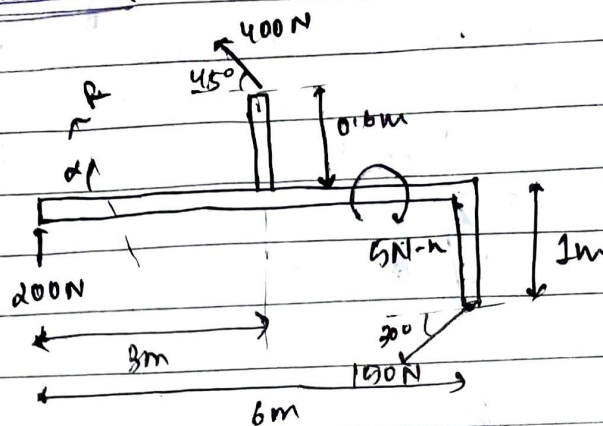
$$R = \sqrt{R_x^2 + R_y^2} = 68.06 \text{ kN}$$

$$\alpha = \tan^{-1} \left(\frac{R_y}{R_x} \right) = 81.55^\circ$$

Taking moment about A

$$R \sin \alpha \times 6 = (20 \times 1.5) + (30 \times 3) + (20 \times 6)$$

Ques:



$$\sum F_x = R_x = (-400 \cos 45 - 150 \cos 30) \text{ N} \leftarrow$$

$$\sum F_y = R_y = (200 + 400 \sin 45 - 150 \sin 30) \text{ N} \uparrow$$

$$R = \sqrt{R_x^2 + R_y^2} = 580.25 \text{ N}$$

$$\alpha = \tan^{-1} \left(\frac{R_y}{R_x} \right) = 44.65^\circ$$

Taking moment about A

$$(400 \cos 45 \times 0.6) + (400 \sin 45 \times 3) + (-150 \cos 30 \times 1) + (-150 \sin 30 \times 6) + (-50) = R \sin \alpha \times 6$$

Types of Beam

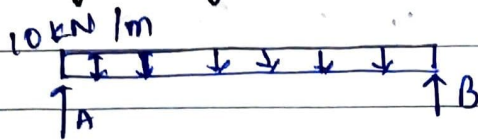
- Simple Supported Beam
- Cantilever Beam
- Fixed Beam
- Overhanging Beam

Types of load

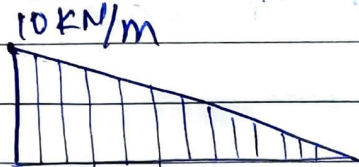
↳ point load / concentrated load



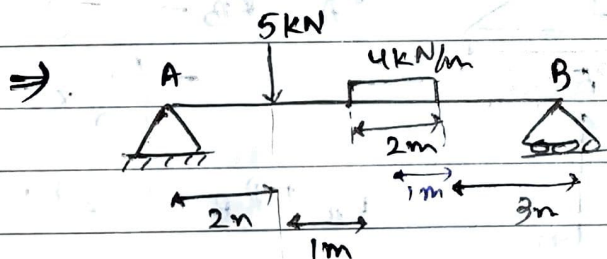
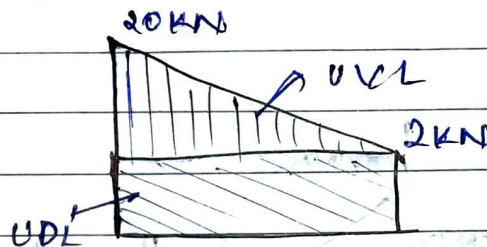
↳ uniformly Distributed load (UDL)



↳ uniformly Varying load (UVL)



Example:-



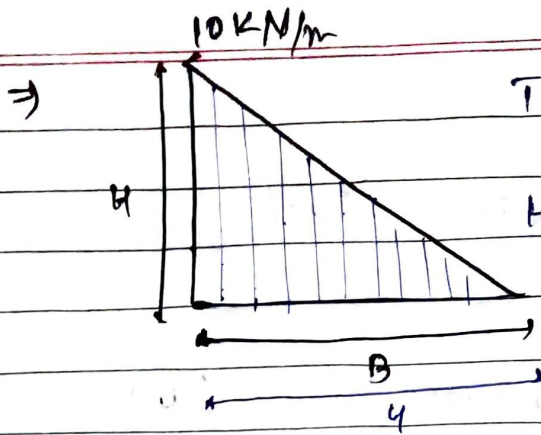
$$\sum F_y = R_y = 5 + (4 \times 2) = 13 \text{ kN}$$

$$R = \sqrt{R_x^2 + R_y^2} = 13 \text{ kN}$$

$$\tan \theta = \infty$$
$$\theta = 90^\circ$$

$$M_B \Rightarrow R \times x = (5 \times 6) + (8 \times 4)$$

$$x = \frac{62}{13} = 4.769 \text{ m}$$

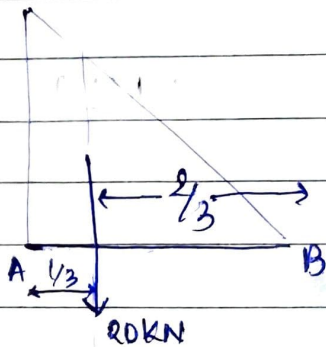


Total load = Area of Δ

Height of the Δ is the max intensity of force

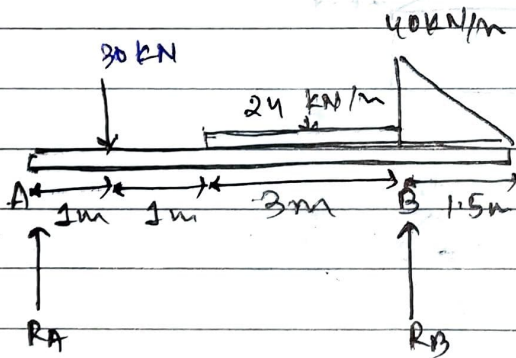
$$H = 10$$

$$\text{Total load} = \frac{1}{2} \times 4 \times 10 = 20 \text{ kN}$$



20 kN will act at the C.G. of the triangle

Ans.



$$R_A + R_B - 30 - (24 \times 3)$$

$$- \left(\frac{1}{2} \times 1.5 \times 40 \right) = 0$$

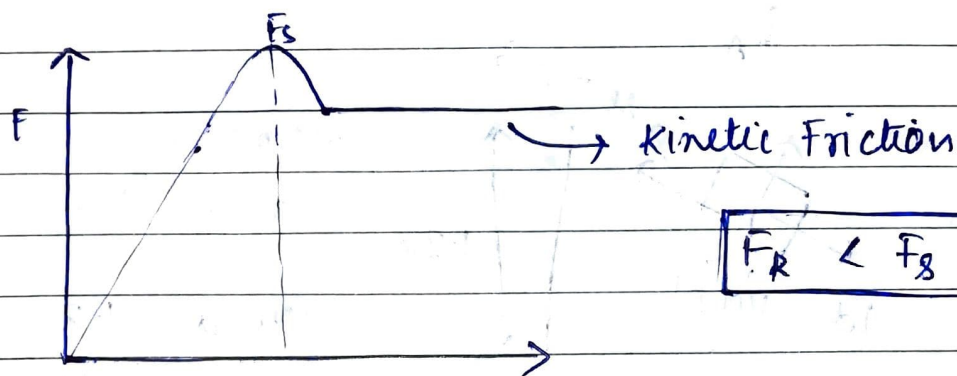
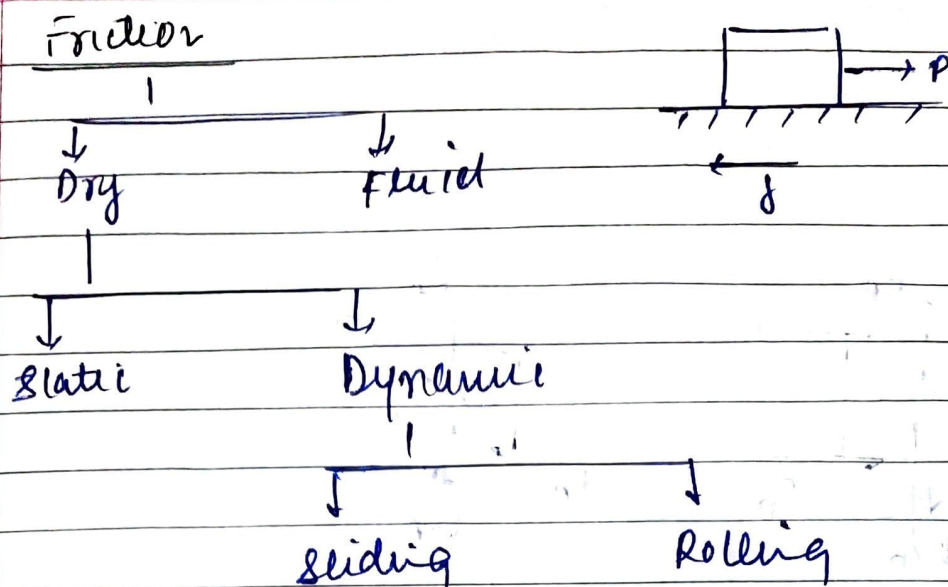
$$R_A + R_B = 132 \text{ kN}$$

$$M_B \Rightarrow (30 \times 4) - (R_A \times 5) + (72 \times 1.5) - \left(30 \times \frac{1}{3} \times 1.5 \right) = 0$$

$$5R_A = 120 + 108 - 15$$

$$R_A = \frac{213}{5} = 42.6 \text{ kN}$$

$$R_B = 89.4 \text{ kN}$$



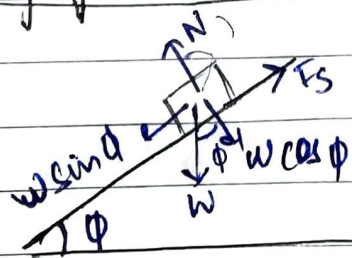
$$F_k < F_s$$

$$\frac{F_s}{N} = \text{constant} = \mu_s \text{ (coefficient of limiting friction)}$$

$$\frac{F_k}{N} = \mu_k \text{ (coefficient of kinetic friction)}$$

$$\mu_s > \mu_k$$

Angle of friction and angle of Repose



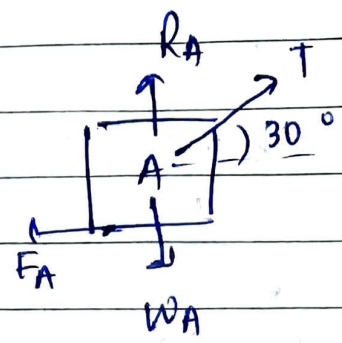
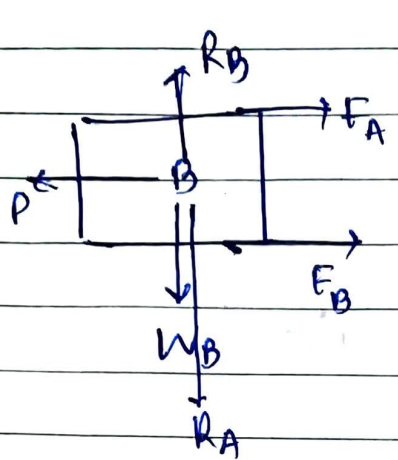
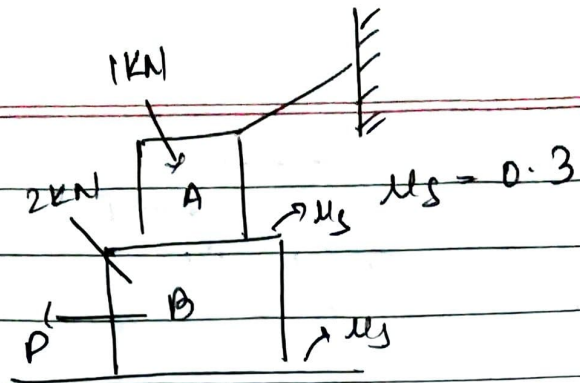
$$w \sin \phi = F_s$$

$$w \cos \phi = N$$

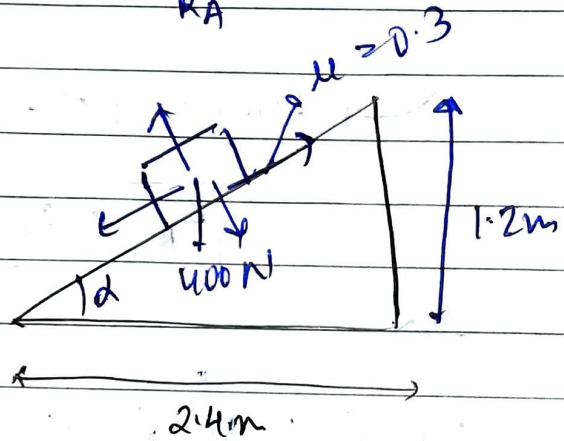
$$\frac{F_s}{N} = \mu_s$$

$$\tan \phi = \mu_s$$

Ques



Ques

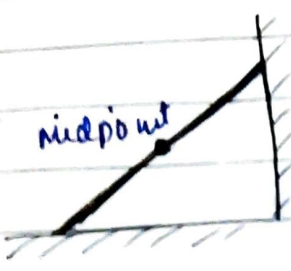


what is required pushing or pulling?

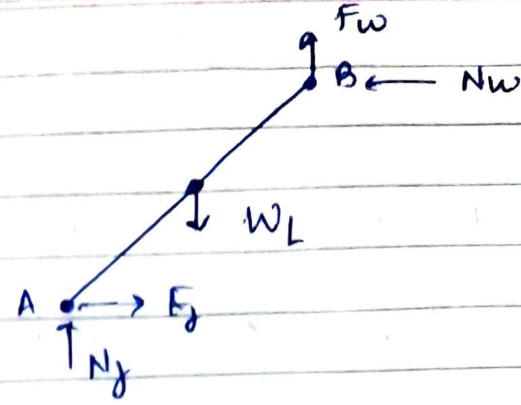
$$\tan \alpha = \frac{1.2}{2.4} = \frac{1}{2}$$

$$400 \times \sin$$

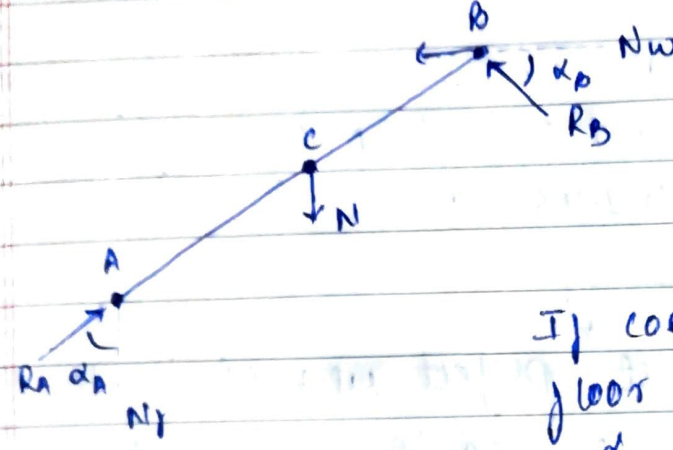
Ladder friction



1.



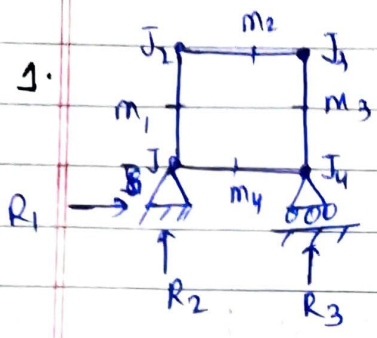
2.



If coefficient of friction of wall
floor & wall is same then
 $\alpha_A = \alpha_B$

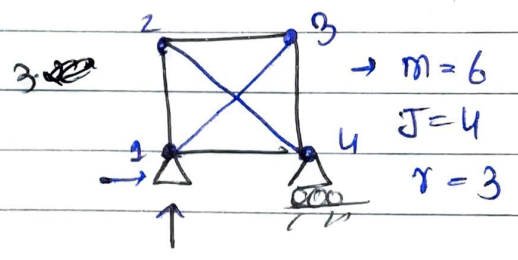
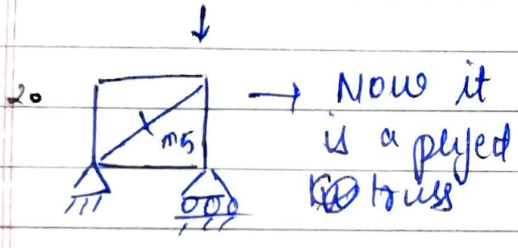
Truss → supports ~~structure~~
↳ sustain's ~~load~~.

- Perfect
 - deficient
 - Redundant
- $m = 2J - r$ → No of reactions
 ↓
 members Joints



$4 = 2 \times 4 - 3$
 $4 \neq 5 \rightarrow$ means deficient
 when L.H.S < R.H.S

now to make it perfect we need to do R.H.S +
so which means we need to increase a member

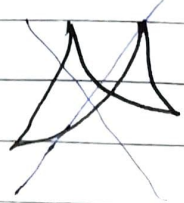


when L.H.S > R.H.S means the truss is redundant which means we need to remove one member like (2).

Truss

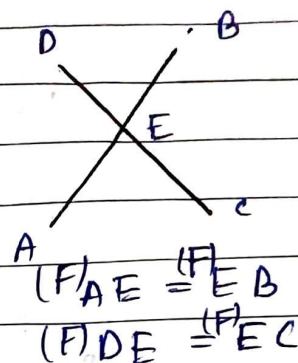
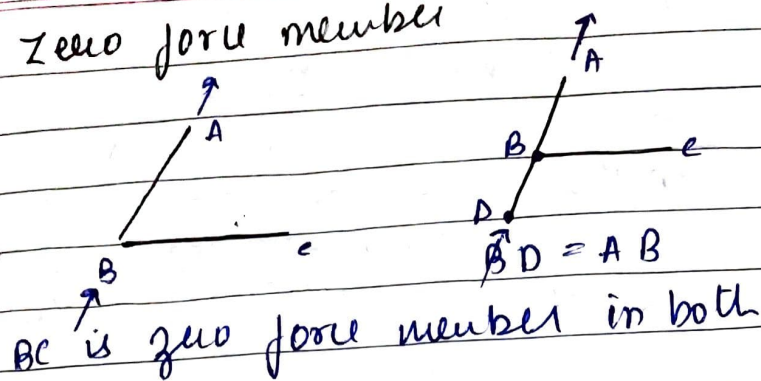
1. Rigid
2. Axial forces only
3. Self weight = 0
4. Forces ~~act~~ that are acting only at the joint
5. Joints are frictionless
6. Members are straight

~~structure~~ → basically just locking & just note that

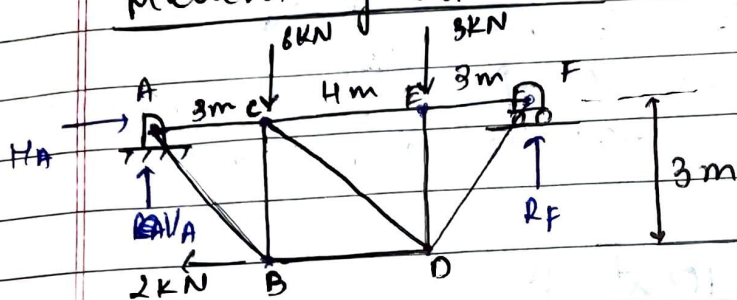


Aisa nhi nota.

Zero force member



Method of Joints



checking whether it is stable or not.

$$m = 2J - 3$$

$$q = 2 \times 6 - 3$$

$$q = 9 \quad \checkmark$$

hence perfect truss.

1. $\sum (M_A = 0 \quad \uparrow)$

$$6 \times 3 + 3 \times 7 + 2 \times 3 - R_F \times 10 = 0$$

$$\frac{18 + 21 + 6}{10} = R_F = 4.5 \text{ kN} (\uparrow)$$

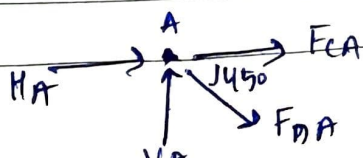
2. $\sum F_x = 0 \Rightarrow H_A - 2 = 0 \Rightarrow H_A = 2 \text{ kN} (\rightarrow)$

3. $\sum F_y = 0 \Rightarrow 6 + 3 = R_F + V_A \Rightarrow 9 = 4.5 + V_A$
 $V_A = 4.5 \text{ kN} (\uparrow)$

~~No we 1st considering joint A~~

* Now we need to consider a joint with minimum (preferably 2) no of members attached or min force acting at that joint

So 1st considering joint A



$$\sum F_x = 0$$

$$H_A + F_{CA} + F_{BA} \cos 45 = 0$$

$$F_{CA} + F_{BA} \cos 45 = -2 \quad \text{--- (1)}$$

$$\sum F_y = 0 \Rightarrow V_A - F_{BA} \sin 45^\circ = 0$$

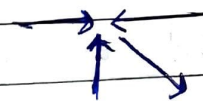
$$V_A = F_{BA} \sin 45^\circ$$

$$F_{BA} = \frac{4.5}{\sin 45^\circ} = 6.36 \text{ kN (T)}$$

sin 45° +ve hence it is tension / tensile force

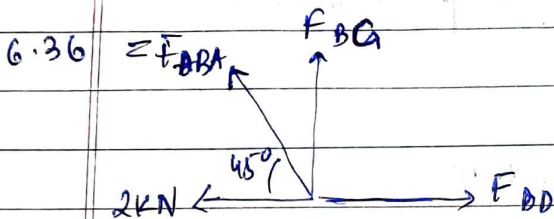
Now putting in (i)

$$F_{CA} = -6.5 \text{ kN (C)}$$



Since -ve hence compressive, so assumed dirⁿ be opposite ~~the~~ dirⁿ mai lagegi.

Now considering joint B



$$\sum F_x = 0 \Rightarrow -2 + F_{AD} - 6.36 \cos 45^\circ = 0$$

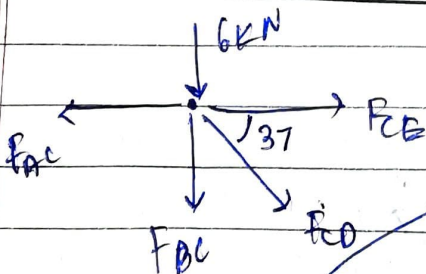
$$F_{AD} = 6.5 \text{ kN (T)}$$

$$\sum F_y = 0$$

$$F_{BC} + 6.36 \sin 45^\circ = 0$$

$$F_{BC} = -4.5 \text{ kN (C)}$$

Now Joint C



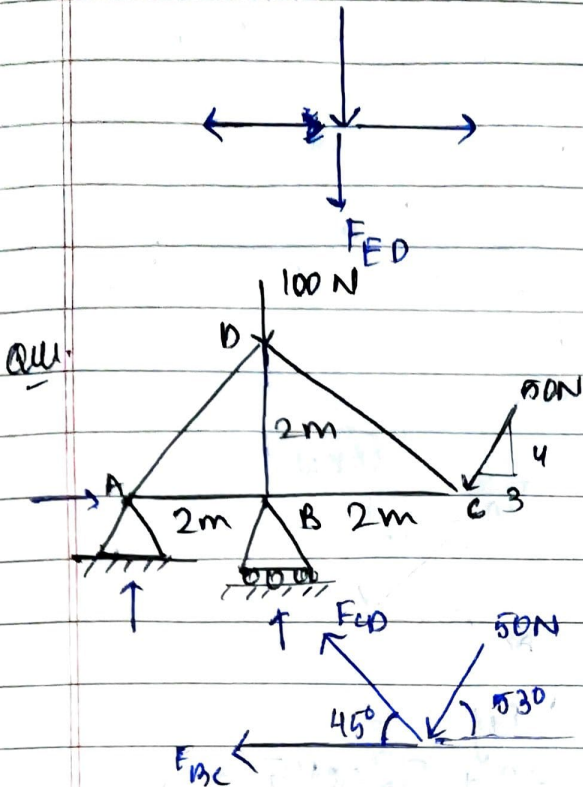
$$F_{CE} + F_{CD} \cos 37^\circ - 6.5 = 0$$

$$F_{CE} = 4.5 \text{ kN (C)}$$

$$-6 - F_{BC} - F_{CD} \sin 37^\circ = 0$$

$$F_{CD} = 2.5 \text{ kN (C)}$$

Now taking Joint E



- 1 → C
- 2 → D
- 3 → B
- 4 → A

$$\sum F_x = 0$$

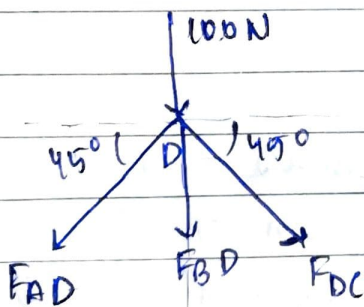
$$-F_{DC} - F_{ED} \cos 45 - 50 \cos 53 = 0$$

$$F_{DC} = 56.568 \times \cos 45 - 50 \cos 53$$

$$\sum F_y = 0 \rightarrow F_{ED} \sin 45 - 50 \sin 53 = 0$$

$$F_{ED} = \frac{50 \sin 53}{\sin 45} = 56.568$$

$$F_{DC} = -70 \text{ (C)}$$



$$\sum F_x = 0$$

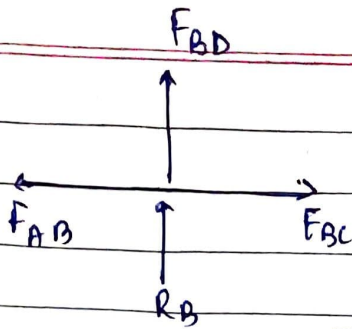
$$-F_{AD} \cos 45 + F_{DC} \cos 45 = 0$$

$$F_{AD} = \frac{F_{DC} \cos 45}{\cos 45} = F_{DC} \text{ (T)}$$

$$\sum F_y = 0 \Rightarrow -100 - F_{DC}$$

$$-F_{BD} - F_{AD} \sin 45 - F_{DC} \sin 45 - 100 = 0$$

$$F_{BD} = -180 \text{ (C)}$$

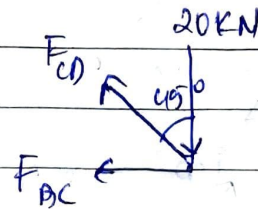
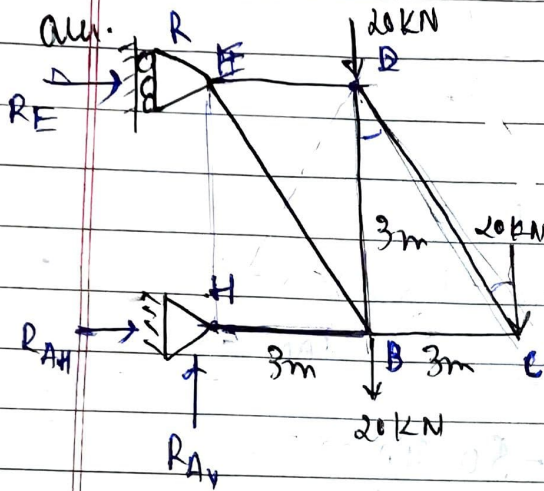


$$\begin{aligned} -F_{AD} - R_B &= 0 \\ R_B &= -F_{AD} \\ &= -(-180) \\ &= 180 \end{aligned}$$

$$-F_{AB} + F_{BC} = 0$$

$$F_{BC} = F_{AB}$$

$$F_{AB} = -70 \text{ (C)}$$

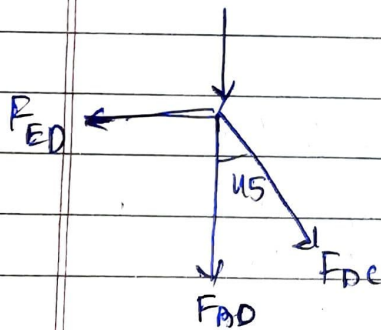


$$\begin{aligned} \sum F_y &= 0 \\ -20 + F_{CD} \cos 45 &= 0 \end{aligned}$$

$$F_{CD} = \frac{20}{\cos 45} = \frac{20}{0.7071} = 28.284 \text{ (T)}$$

$$\sum F_x = 0 \Rightarrow -F_{BC} - F_{CD} \sin 45 = 0$$

$$\begin{aligned} F_{BC} &= -28.284 \times \sin 45 \\ &= -20 \text{ kN} \end{aligned}$$



$$\sum F_x = 0$$

$$-F_{ED} + F_{DC} \sin 45 = 0$$

$$F_{ED} = F_{DC} \sin 45 = 20 \text{ (T)}$$

$$\sum F_y = 0 \Rightarrow -20 - F_{BD} - F_{CD} \cos 45 = 0$$

$$F_{BD} = -20 \text{ (C)}$$

$$R_{AH} + R_E = 0$$

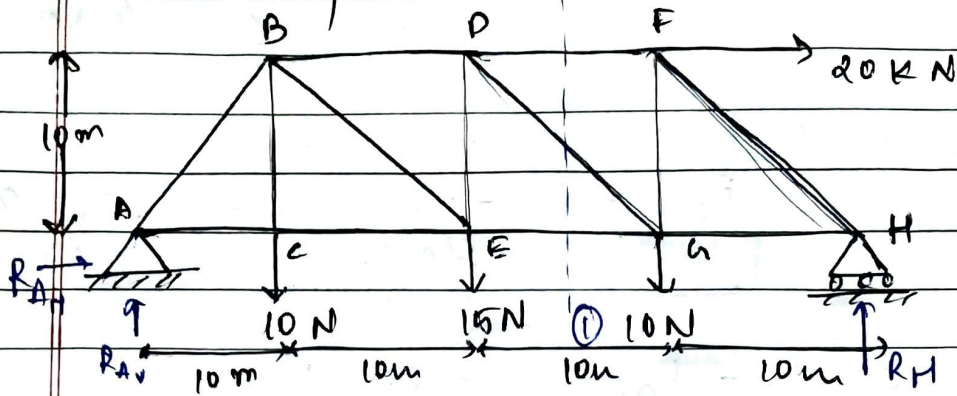
$$\begin{aligned} R_{AV} &= 20 + 20 + 20 \\ &= 60 \text{ kN} \end{aligned}$$

$$\sum M_H = 0 \Rightarrow 20 \times 3 + 20 \times 6 + 20 \times 3 + R_E \times 3 = 0$$

$$R_E \times 3 = 60 + 120 + 60$$

$$= -240 \Rightarrow R_E = -80$$

Method of Section ①



Find forces in DF, DH, EG

★ First find support reactions

$$\sum F_x = 0 \Rightarrow R_{AH} + 20 = 0 \Rightarrow \boxed{R_{AH} = -20}$$

$$\sum F_y = 0 \Rightarrow R_{AV} + R_H - 10 - 15 - 10 = 0$$

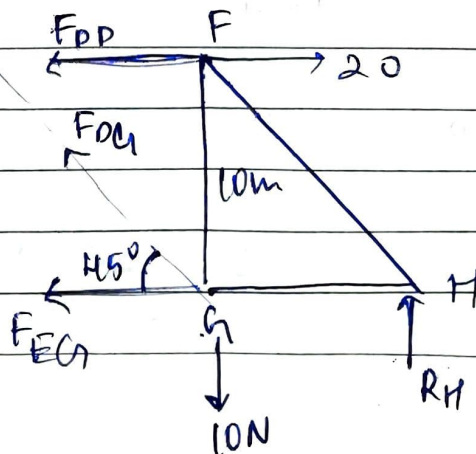
$$R_{AV} = 12.5$$

$$\sum M_A = 0 \Rightarrow 10 \times 10 + 15 \times 20 + 10 \times 30 - R_H \times 40 + 20 \times 10 = 0$$

$$\boxed{R_H = 22.5 \text{ N}}$$

★ Now, Jab bhi ap truss ko sectional plane se cut krte hai to maximum 3 unknown member cut kr chahiye.

Now cutting through section 1-1 and draw that side (LHS/RHS) with or FBD of smaller side



Now we have to apply equilibrium condⁿ
use that eqⁿ which has only 1 unknown

$$\sum F_y = 0 \Rightarrow +F_{DG} \sin 45 - 10 + R_H = 0$$

$$\underline{F_{DG} = \frac{10 - 22.5}{\sin 45} = -17.678 \text{ (C)}} \quad (C)$$

$$\sum M_A = 0 \Rightarrow 20 \times 10 - F_{DE} \times 10 - R_H \times 10 = 0$$

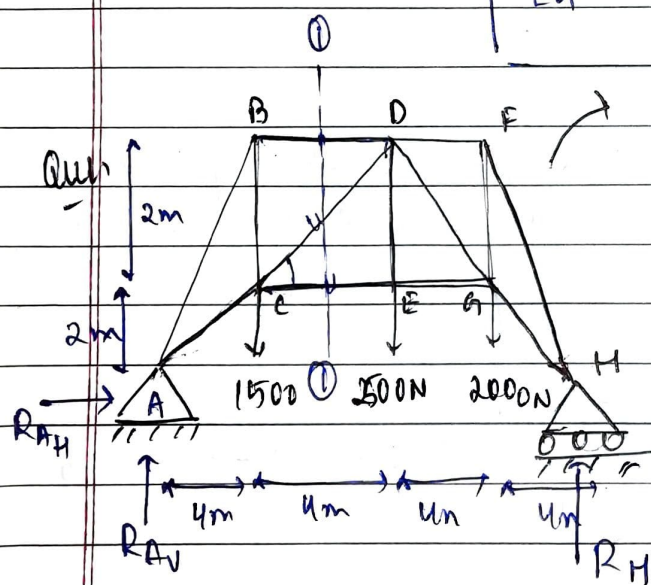
$$F_{DE} \times 10 = 200 - 22.5 \times 10$$

$$\underline{F_{DE} = -2.5 \text{ (C)}} \quad (C)$$

$$\sum F_x = 0 - F_{EG} - F_{DE} - F_{DG} \cos 45 + 20 = 0$$

$$F_{EG} = +2.5 + 17.677 \times \cos 45 + 20$$

$$\underline{F_{EG} = 35 \text{ N (T)}} \quad (T)$$



BD, CD, CE joints

$$\sum F_x = 0 \quad R_H = 0$$

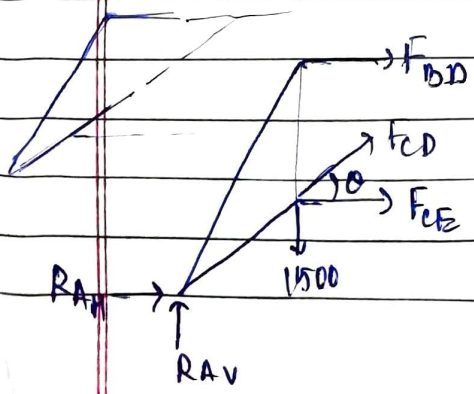
$$\sum F_y = 0 \quad R_{AV} + R_H = 6000$$

$$\underline{R_{AV} = 2875 \text{ N}}$$

$$\sum M_A = 0$$

$$1500 \times 4 + 2500 \times 8 + 2000 \times 12 - R_H \times 16 = 0$$

$$\underline{R_H = 3125 \text{ N}}$$



$$\tan \theta = \frac{2}{4} = \frac{1}{2} \Rightarrow \theta = \tan^{-1}(\frac{1}{2})$$

$$= 26$$

$$\theta = \tan^{-1}(\frac{1}{2})$$

$$= 20.565$$

$$\sum F_y = 0 \Rightarrow F_{CD} \sin 26.565 - 1500 + 2875 = 0$$

$$F_{CD} = -3074.598 \text{ N (C)}$$

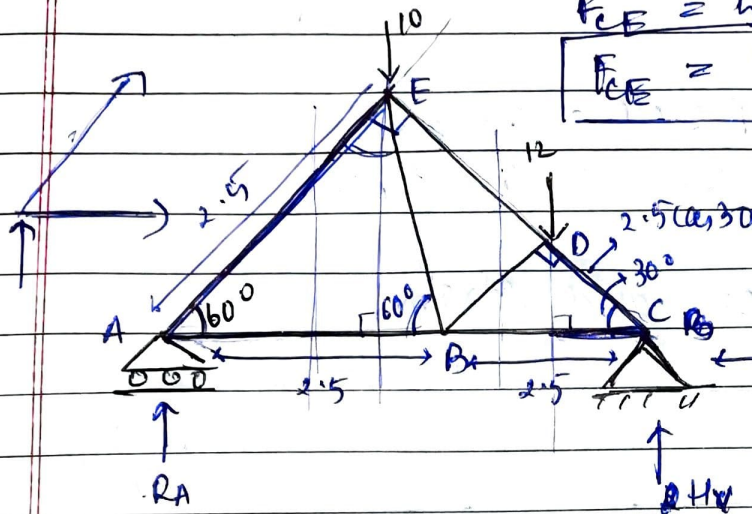
$$\sum M_c = 0 \Rightarrow F_{BD} \times 2 + R_{AD} \times 4 = 0$$

$$F_{BD} = \frac{-2875 \times 2}{2} = -5750 \text{ N} \quad (c)$$

$$\sum F_x = 0 \Rightarrow F_{BD} + F_{CD} \cos 26.565 + F_{CE} = 0$$

$$F_{EF} = 5750 + 3074 \cdot 598 \times \cos 26.85^\circ$$

$$F_{CE} = 8500 \text{ N (T)}$$



$$\sum F_x = 0$$

$$H_c = 0$$

- H.C $\sum_i F_y = 0$

$$H_v + R_A - 10 - 12 = 0$$

$$F_{by} + R_A = 22$$

$$\sum_i M_i = 0$$

Ans 30 =

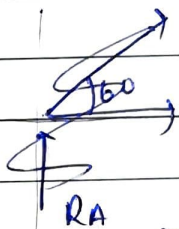
~~$\cos 30^\circ = DC$~~

$$\tau_{12} = -10 \times 2.5 \cos 30$$

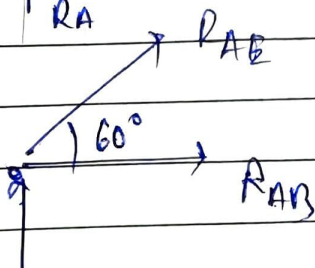
$$-12 \times 2.5 \cos 30 \cos 30$$

~~$$DL = 2.5 \times \sqrt{3}$$~~

$$+ R_A(5) = 0$$



$$R_A(5) = 10 \times 2.5 \cos 30 + 12 \times 2.5 \cos 30 \times \cos 30$$



$$R_{AB} + R_{AE} \text{ (MGO)} = 6$$

$$R'_A + R_{AB} \sin 60 = 0$$

